

Project Title

Graph Enhanced Bayesian Network for Explainable Financial Risk Prediction

Abstract

Financial institutions increasingly require accurate and interpretable models to assess credit and financial risk in complex, data rich environments. Traditional machine learning methods often achieve high predictive performance but provide limited interpretability, while probabilistic models may struggle to capture hidden relationships among financial entities. This project proposes a hybrid framework that integrates Graph Neural Networks (GNNs) with Bayesian Networks to improve financial risk prediction through relational representation learning and probabilistic reasoning. The methodology begins with preprocessing financial tabular data and constructing a financial relationship graph based on feature similarity or domain specific connections. A GNN is then employed to learn informative node embeddings that capture latent interactions among customers or transactions. These learned representations are discretized and provided as inputs to a Bayesian Network, which performs explainable probabilistic inference to estimate financial risk and confidence levels. The proposed framework aims to combine the representation learning capability of deep learning with the interpretability and uncertainty handling of probabilistic graphical models. The project belongs to the domains of Artificial Intelligence, Machine Learning, Graph Machine Learning, Explainable Artificial Intelligence, and Financial Technology. Its primary applications include credit risk assessment, loan default prediction, fraud detection, and financial decision support. The motivation for employing probabilistic reasoning models lies in their ability to represent conditional dependencies, quantify uncertainty, support transparent decision making, and provide confidence aware predictions, making them well suited for high stakes financial applications where reliability and interpretability are essential.